

Estimation of Productivity of Rippers.

Considering the generalised case when the ripping is done in $\parallel^{el}$ and X-passes.

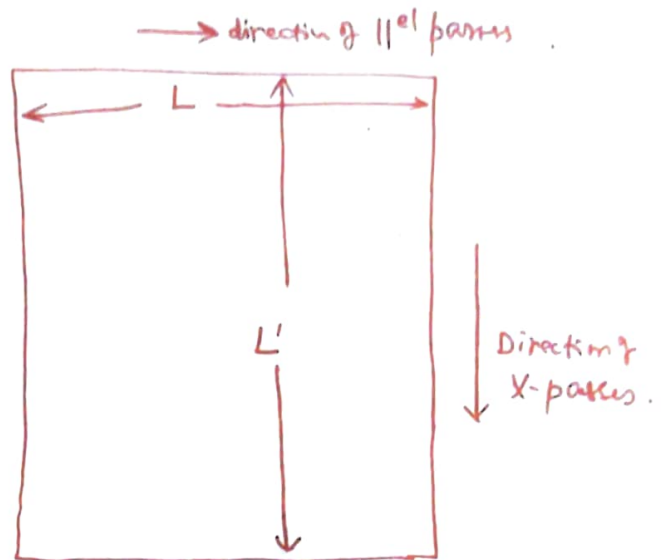
given parameters

V_r = ripping Velocity in top gear. m/s.

L = length of $\parallel^{el}$ passes.

L' = Length of cross-passes.

T = time of ripper to travel back to next furrow. (seconds) during $\parallel^{el}$ pass



~~Assumption $\rightarrow L$ & L' are given that $L' \neq 0$ & $L \neq 0$~~

T' = time of ripper to travel back to next furrow m. (second) during X-passes.

C = ? Spacing between two passes in $\parallel^{el}$

C' = ? Spacing between two passes in X-direction.

Now:

No of passes required in $\parallel^{el}$ cut = $\left(\frac{L'}{C}\right)$.

No of passes required in X-cut = $\left(\frac{L}{C'}\right)$.

$$\text{Total time in } \parallel^{el} \text{ pass} = \left[\frac{L'}{C} \times \frac{L}{V_r} + T \cdot \frac{L'}{C} \right]$$

$$\text{Total time in X-pass} = \left[\frac{L}{C'} \times \frac{L'}{V_r} + T' \cdot \frac{L}{C'} \right]$$

$$\begin{aligned} \therefore \text{Total time for both } \parallel^{el} \& \text{ X-pass} &= \left[\frac{LL'}{C V_r} + \frac{T L'}{C} + \frac{LL'}{C' V_r} + \frac{T' L}{C'} \right] \\ &= \left[\frac{LL'}{V_r} \left(\frac{1}{C} + \frac{1}{C'} \right) + \frac{T L'}{C} + \frac{T' L}{C'} \right] \end{aligned}$$

(2).

Now total volume excavated in 11' & X-pit

$$= h_t \times L \times L' \quad m^3.$$

$$\therefore \text{Theoretical productivity } \left(\frac{m^3}{s} \right) = \frac{h_t \times L L'}{\left[\frac{L L'}{v_r} \left(\frac{1}{c} + \frac{1}{c'} \right) + T \left(\frac{L'}{c} \right) + \frac{T' L}{c'} \right]}$$

If L' is chosen such that $\odot (T = T')$

Then.

$$\text{Theoretical prodn } \left(\frac{m^3}{s} \right) = \frac{h_t \cdot L L'}{\frac{L L'}{v_r} \left(\frac{1}{c} + \frac{1}{c'} \right) + \left(\frac{T L'}{c} + \frac{T L}{c'} \right)}$$

$$= \frac{h_t}{\frac{1}{v_r} \left(\frac{1}{c} + \frac{1}{c'} \right) + T \left(\frac{1}{L c} + \frac{1}{L' c'} \right)}$$

So
practical hourly
productivity considering K_u (ripper utilization factor)
(0.7 to 0.8)

$$Q_r = \frac{3600 K_u \times h_t}{\frac{1}{v_r} \left(\frac{1}{c} + \frac{1}{c'} \right) + T \left(\frac{1}{L c} + \frac{1}{L' c'} \right)} \quad \frac{m^3}{hr}.$$

(3).

Productivity Calculation for 11^{el} parts only.

$$\text{No of parts in } 11^{\text{el}} = \frac{L'}{C}$$

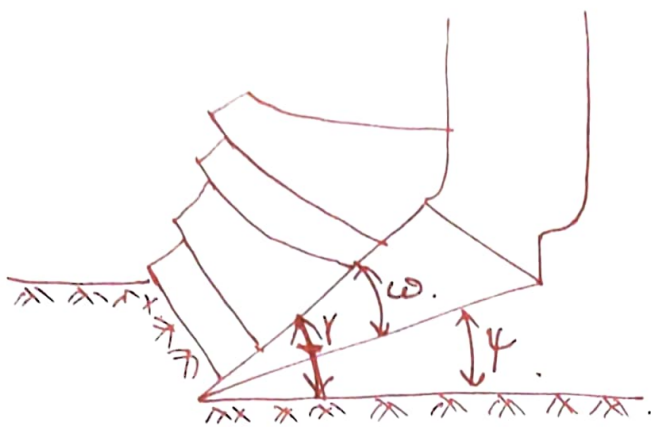
$$\text{Volume loosened in } 11^{\text{el}} \text{ parts} = (h_{\text{eff}} \times L \times L')$$

$$\text{Total time taken in } 11^{\text{el}} \text{ parts} = \left(\frac{L'}{C}\right) \frac{L}{V_r} + \frac{T L'}{C}$$

$$\begin{aligned} Q \left(\frac{\text{m}^3}{\text{s}} \right) &= \frac{h_{\text{eff}} \times L L'}{\frac{L L'}{C V_r} + \frac{T L'}{C}} \\ &= \frac{h_{\text{eff}}}{\frac{1}{C V_r} + \frac{T}{L C}} = \frac{h_{\text{eff}} \times C}{\left(\frac{1}{V_r} + \frac{T}{L}\right)} \end{aligned}$$

$\therefore$ Hourly productivity considering effective ripper utilization (0.7).

$$Q_r (11^{\text{el}}) = \left(\frac{3600 \times C \times h_{\text{eff}} \times K_u}{\frac{1}{V_r} + \frac{T}{L}} \right)$$



γ = cutting angle.

ω = tip angle

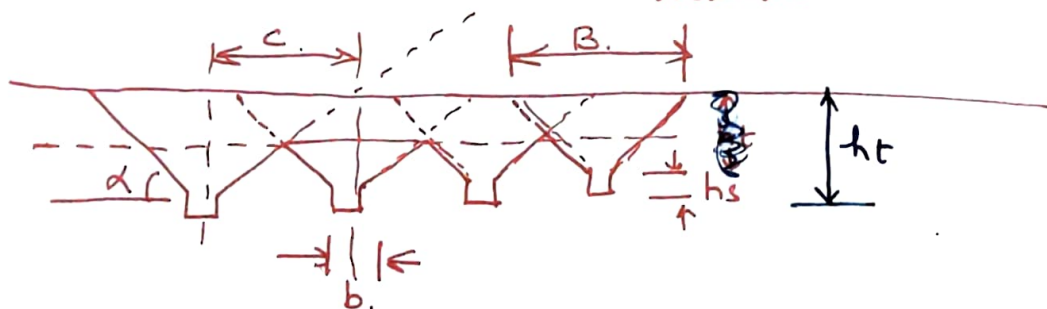
ψ = tooth clearance angle.

$$\omega = 20 \text{ to } 30^\circ$$

$$\phi = 5 \text{ to } 7^\circ \text{ (igneous/metamorphic rock)}$$

$$\psi = 8 \text{ to } 10^\circ = \text{frozen rock mass}$$

γ is automatically defined from $\psi + \omega$ for a smaller clearance angle causes crushing of rock mass.



Slot width = b . (close to thickness of tooth tip (b_t)).

$$h_s = \text{slot height} \cdot (0.15 \text{ to } 0.2 \text{ times ripper tooth penetration } h_t)$$

$$\alpha = \text{slope of sidewall of the cut } (40-60^\circ)$$

depends on: rock mass characteristics & tooth tip geometry

rock mass loosening ability of a ripper is specified by

— possible penetration (h_t) of ripper tooth into the loosened rock mass

monolithic rock mass → loosening due to overcoming tensile stress

jointed rock mass → overcoming cohesion bet str. blocks.

C = distance between adjacent passes.

— set with an aim to ensure required lump size.

— sufficient loosening depth.

Rippers

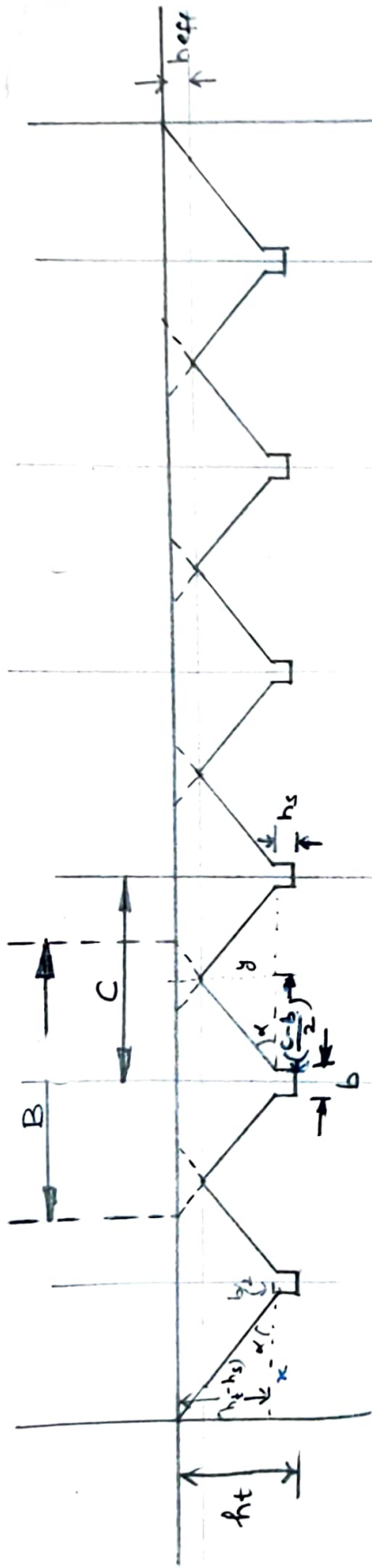


Figure :- for 1st cuts made by ripper.

$$\frac{h_t - h_s}{x} = \tan \alpha$$

$$\Rightarrow x = (h_t - h_s) \cot \alpha$$

$$\frac{y}{(\frac{C-b}{2})} = \tan \alpha$$

$$y = (\frac{C-b}{2}) \tan \alpha$$

$$\frac{y}{(\frac{C-b}{2})} = \tan \alpha$$

B = Width of furrow

C = distance between adjacent furrows

Considering n-parallel furrows, The total Volume of material

$$V = [(n-1)C + B] h_{eff} \quad (1)$$

$$h_{eff} = h_t - h_s - y$$

$$= (h_t - h_s) - (\frac{C-b}{2}) \tan \alpha \quad (2)$$

Not used directly

$$\frac{B}{2} = (h_t - h_s) \cot \alpha + \frac{b}{2}$$

$$\Rightarrow B = 2(h_t - h_s) \cot \alpha + b$$

(2)

Using Equation (1) & (2) we get

$$V = [(n-1)C + B] \left[(h_t - h_s) - \left(\frac{C-b}{2} \right) \tan \alpha \right]$$

Now expanding we get

$$V = (n-1)(h_t - h_s)C - \frac{(n-1)C(C-b)}{2} \tan \alpha + B(h_t - h_s) - B \times \left(\frac{C-b}{2} \right) \tan \alpha$$

$$\text{or } V = (n-1)(h_t - h_s)C - \frac{(n-1)C^2}{2} \tan \alpha + \frac{(n-1)bC}{2} \tan \alpha + B(h_t - h_s) - \frac{Bc \tan \alpha}{2} + \frac{bB}{2} \tan \alpha$$

differentiating

$$\frac{dV}{dC} = (n-1)(h_t - h_s) - (n-1)C \tan \alpha + \frac{(n-1)b \tan \alpha}{2} - \frac{B \tan \alpha}{2} = 0$$

$$\Rightarrow (n-1)(h_t - h_s) - (n-1)C_{\text{opt}} \tan \alpha + \frac{(n-1)b \tan \alpha}{2} - \frac{B \tan \alpha}{2} = 0$$

$$\text{or, } (n-1)(h_t - h_s) + \frac{(n-1)b \tan \alpha}{2} - \frac{B \tan \alpha}{2} = (n-1)C_{\text{opt}} \tan \alpha$$

$$\text{or, } C_{\text{opt}} \times [n-1] \tan \alpha = (n-1)(h_t - h_s) + \frac{(n-1)b \tan \alpha}{2} - \frac{B \tan \alpha}{2}$$

$$\Rightarrow C_{\text{opt}} = (h_t - h_s) \cot \alpha + \frac{b}{2} - \frac{B}{2} \times \frac{1}{(n-1)}$$

$$\text{Now as } n \rightarrow \infty \Rightarrow \frac{B}{2} \times \frac{1}{n-1} \rightarrow 0$$

$$\therefore C_{\text{opt}} = (h_t - h_s) \cot \alpha + \frac{b}{2}$$

(3)

Now Using this value of C_{opt} in Equation (2) we get.

$$h_{eff} = (h_t - h_s) - \frac{(h_t - h_s) \cot \alpha + \frac{b}{2} - b}{2} \tan \alpha$$

$$= (h_t - h_s) - \frac{(h_t - h_s) \cot \alpha + \frac{b}{2} - b}{2} \tan \alpha$$

$$= (h_t - h_s) - \frac{(h_t - h_s) + \frac{b}{2} \tan \alpha}{2}$$

$$= \frac{h_t - h_s}{2} + \frac{b}{4} \tan \alpha$$

$$\therefore h_{eff} = \frac{(h_t - h_s)}{2} + \frac{b}{4} \tan \alpha$$